

3-merkle_github_anchor.py

User Instructions

Overview

3-merkle_github_anchor.py is used to publish a generated manufacturing batch to a third-party public record.

The program uploads a previously generated:

```
batch_<facility_id>_<instrument_id>_<unique_id>.json
```

file to GitHub Gist.

This creates a publicly accessible timestamped reference for the manufacturing batch.

The purpose of anchoring is accountability rather than prevention.

Launching the Program

Run the script from a terminal:

```
python 3-merkle_github_anchor.py
```

A Tkinter graphical interface window will appear.

Purpose of Public Anchoring

The manufacturing batch generated by:

```
2-merkle_batch_generator.py
```

already contains: - cryptographic job hashes - Merkle proofs - manufacturing references

However, the batch still exists only on the local machine.

Uploading the batch to a third-party public location creates: - a timestamped public reference - external proof the batch existed at a certain time - evidence against silent deletion or modification

Why GitHub Gist?

GitHub Gist is used in this demonstration because it is: - publicly accessible - easy to use - widely trusted - free - timestamped - simple to integrate

This demonstration uses GitHub primarily as a: - proof-of-existence layer - public registry - temporary anchoring mechanism

The project could later be adapted to: - blockchain systems - distributed ledgers - institutional registries - dedicated audit servers

GitHub Gist is used here because it lowers the barrier to experimentation and public exploration.

Interface Sections

The interface contains several sections used to select and upload a manufacturing batch.

Batch File Selection

Use the:

Select Batch File

button to choose a previously generated batch JSON file.

Example:

batch_ForgeWorks_Printer01_9f2a1c4d.json

This file is normally created by:

2-merkle_batch_generator.py

GitHub Token Selection

Use the:

Select GitHub Token

button to optionally select a:

github_token.txt

file.

This token allows the script to authenticate with GitHub.

The token is used only for: - creating the Gist - uploading the batch file

The token itself is NOT uploaded publicly.

Why Use a Token?

GitHub requires authentication for creating Gists through the API.

The token allows: - automated uploads - authenticated requests - ownership of the created Gists

Without a token, the script cannot communicate with the GitHub API.

Private Option

The:

`Private`

checkbox determines whether the created GitHub Gist is publicly searchable.

Unchecked

The Gist is public.

Anyone with the link can: - view the batch - download the batch - inspect the Merkle structure

This is useful for: - demonstrations - public accountability - collaborative development

Checked

The Gist becomes unlisted.

People with the direct link can still access it, but it is not publicly indexed.

This is useful for: - internal testing - limited demonstrations - semi-private audit workflows

Input Preview Window

The input preview displays the selected batch JSON file before upload.

This allows the user to: - inspect the batch contents - confirm the correct file was selected - review metadata before publication

The preview may include: - batch ID - Merkle root - job count - proofs - descriptions - manufacturing references

Anchor Button

Press:

`Anchor Batch`

to upload the selected batch to GitHub Gist.

When pressed, the program:

1. Reads the selected batch JSON
 2. Authenticates with GitHub
 3. Creates a new Gist
 4. Uploads the batch file
 5. Retrieves the public Gist URL
 6. Displays the resulting public reference
-

Output Preview Window

After upload completes, the output section displays information returned from GitHub.

This may include: - Gist URL - Gist ID - upload status - public/private status - API response details

This section serves as confirmation that the batch was successfully anchored.

What the Public Anchor Represents

The uploaded Gist does NOT prove: - the legality of the object - the intent of the operator - the safety of the manufactured item

Instead, it demonstrates: - that a manufacturing batch existed - when it existed - what cryptographic records were associated with it

The anchor functions as: - an accountability reference - a tamper-evidence mechanism - a proof-of-existence record

Recommended Workflow

Typical usage:

1. Generate manufacturing jobs using:

```
1-merkle_integrated_slicer.py
```

2. Generate a manufacturing batch using:

```
2-merkle_batch_generator.py
```

3. Open:

```
3-merkle_github_anchor.py
```

4. Select:

- batch JSON file
- GitHub token file

5. Optionally select:

```
Private
```

6. Press:

```
Anchor Batch
```

7. Review the returned GitHub Gist information
-

Purpose of the Demonstration

This program demonstrates: - public proof-of-existence anchoring - decentralized accountability concepts - external timestamping - append-only manufacturing records - public audit references

The GitHub anchoring step is intentionally lightweight and accessible.

It is intended to encourage: - experimentation - public exploration - collaborative development - discussion of voluntary accountability systems